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ABSTRACT

The Smart Energy Harvesting using physical exercising machines which are feasible enough to harvest power seems to generate a new source of renewable energy. Biomechanical energy harvesting constitutes a clean, portable energy alternative to the fossil fuels currently being used. Biomechanical energy harvesting is a novel approach for producing energy for small devices. The energy expended in a typical workout at the gym is usually wasted in the mechanics of the equipment. This project will harness the mechanical energy of the machine and convert it to electrical energy using a generator based system. The exercise equipment will be attached to a generator. This creates a dc voltage which will be fed into a DC-DC converter and then sent to a battery where it can be stored for future use. This proposed system will help consuming our physical energy in some useful way.

KEYWORDS

Biomechanical energy, Harness, DC-DC converter.

1. INTRODUCTION

The proposed discipline of energy conservation is becoming an increasingly extraordinary subject of study among the scientific neighborhood at present. Brainiacs of all times use their knowledge to extract ways of providing cheap and green energy. The purpose of this venture is to build a human powered generator from a used exercising machine and to use it to power mild lights, mobile telephones and other small appliances. It will aid and strengthen engineering expertise and learning about an easy approach to produce electrical power which can be gained from various mechanical sources.

1.1 Overview

Over the past decade, scientists and engineers around the world had been designing exceptional energy-harvesting programs, drawing energy from a form of sources. One of the vital ingenious and unlimited sources to be had is the kinetic energy produced during human exercise [1]. Even though latest designs of energy-harvesting undertaking apparatus have been offered into the market which can convert the mechanical energy wasted during physical exercise, these systems are expensive , inefficient and do not produce an obvious output of power [2]. These systems need to be improved, expanded and designed for highest energy output, price-affectivity, and marketability. Engineered to be used for retrofitting current undertaking laptop, this venture involves an effective yet controllable energy storage through multiple devices into a single energy storage battery and distribution procedure to various appliances. The objective of this is to create and develop a regenerative energy source based round a piece of undertaking gear [3]. Also, environmentalists who are interested in minimizing environmental impacts due to current energy producing means and people who need to preserve the atmosphere will use and will support this form of electrical power new release thereby lowering the emission of hazardous gases and matter to the surroundings. The energy used in an average exercise on the gym is as a rule lost in the process of the application of the apparatus. The course of action lessens the automated energy and is modified to electrical energy utilizing a generator-based process [4]. Thus, it reduces the power usage coming from the electricity distribution companies and produces eco-friendly and low cost energy. The energy harvesting equipment might be attached to the channel of the generator based system [5]. Thus, producing electrical energy used in powering a piece of gear such as lamp, bulb or laptop while exercising [6].

2. LITERATURE REVIEW

Energy harvesting systems relies on several groups of devices/machines which can convert the waste energy into usable energy. The groups of the devices which function in these systems are piezoelectric crystals, dynamos/generators, solar cells and other thermoelectric generators.

Piezoelectric convert the mechanical stress produced into the electrical energy. This can be seen in many devices used in our everyday life. This type of effect can be seen in many materials which have the ability to produce electric potential. This effect can be seen in Quartz clocks and cigarette lighters.

Dynamos and generators both have nearly same kind of functionality. They convert the mechanical (rotational) energy into electrical energy by rotating a magnet within the several loops of wire thus producing electromagnetic effect within. This can also happen if the wire is rotated keeping the magnet stationary. Such device and machines can already be seen in wind turbines and hydro electric motors.

Electromagnetic collectors are the machines that convert the solar energy into electrical energy thus used as solar cells which converts the energy coming from the sunlight directly into electrical energy. Other types of such systems are the RF collectors which produces electricity through EM spectrum produced.

2.1 Examples of Energy Harvesting

Energy harvesting process can be achieved from various method and devices thus harvesting the waste energy into electrical energy except the use of electromagnetic harvesters. Scientists of all times have deduced different ways of harvesting energy with different levels. There are large number of everyday actions which can be used in energy harvesting which includes walking, running, breathing and even finger or toes movement (keyboard typing).

1. One main example of energy harvesting by walking and running is the use of piezoelectric materials. Such materials have been placed in jogging shoes which produce energy by piezoelectric effect when the foot touches the ground. Harvesting energy through walking has always been a major focus of the scientists and the researchers due to mobile energy requirements.
2. Another focus on harvesting energy through walking or running is by the bending motion around the knees. A “Biomechanical Energy Harvester” was introduced by some students with the help of some professors at Simon Fraser University which generally focus on energy harvesting by knee bending movement. The device weighing 3.5lb was attached with the body part and it can produce up to 5W of energy which is enough to charge 10 cell phones.
3. Similarly many other stationary systems have been introduced which can harvest energy.

Researchers at a well-recognized university MIT introduce the concept of a piezoelectric floor which produces energy when someone walks on it.

The most important thing in the process of energy harvesting through mechanical systems is the amount of power applied on the machine. The action applied on the machine should be less resistive for the user. According to the law of energy and keeping in and the energy balance of a human body, the maximum energy harvested cannot exceed the amount of energy applied. Thus if a large amount of energy is required then the power exerted on the specified system should be much larger. The energy harvesting system should be efficient enough to offset the initial cost spent on the development of that system and the cost per kilowatt hour (kWh). To make a system more efficient to harvest maximum amount of energy, the harvesting machine may hold up the natural flow of that mechanical motion under which the system is working.

2.2 Prospective of Man Energy

Power consumption of man is viewed on various facts, a gigantic and large number of capabilities appear obvious on observation. For the reason that the ordinary 2000kcal of everyday utilization by a human body (97W of power in, on ordinary), people soak up about 8.368MJ or 2324Wh of energy daily. This is roughly the identical amount of energy saved within the average car battery (2400Wh). Nevertheless, the outflow of energy for original tasks is quite excessive as well as seen in table 2.1. This table show maximum amount of power that may be captured as a consequence of human exercise.

Table 2.1

Action	Energy Used (W)
Pressing Switch	0.63
Squeezing Handle	13
Revolving Crank	29
Racing Bike	>10

For that reason, the existing power which can be taken for little interval of time is truly fairly confined. Exchange simply some of the biggest ability firewood vigor crops in the US would need roughly the populace of 2 NY city metro areas to be using human vigor generating bicycle:

$$\frac{2980MW}{100PerPerson} = 2980000\ People$$

The apparent uselessness of determine suggests why the choice for that reason some distance in human power new release has been restrained to lessen energy functions similar to customer electronics producing 1799 watts for a couple of unit time must be inside the variety of the satisfactory vigor cranes. Recall 1W approach making use of a drive of 1N via a distance of 1m in 1s.For those who lifted 1kilogram that is 9.8N of drive, about 10N, for 1m in 1s that may be 10W.

2.3 Unit Conversion

First hold in mind that Watts and energy are two unique items of dimension that are not able to be instantly modified backward and forward. However when usage of Watt-Hours rather than simply "Watts" then we definitely have a way to transform it to energy. Listed here are the steps: Convert Watt-Hours to Watt Seconds (Joules), then convert Joules to calories, then alter energy with human body effectively element. So for this instance let's assume that you just furnish pedal power to a one hundred W TV for 1hour. In view that 1J is equal to 1W * unit time we do dimensional analysis and get:

100Wh * (3600 s / 1 H) = 360,000 Joules
1 calorie = 4.184 J to transform J to energy
360,000 J / four.184 = 86,042 energy

Whilst you appear on the label of Oreo cookies or different meals objects on the shop, the term "energy" is particularly (k-energy). So we divide by way of 1000 to get 86 calories energy. Presume that your physique provides 24.9% effective energy when riding cycle you divide by way of 0.25 calories burned during brisk walking a 100W television set for 1 hour = 86 / 0.25 = 344 which is ready identical to the energy obtained by eating 1 piece of pizza.

Another military based project carried out by University of Pennsylvania (Dept. Of Electronics Engineering) was the production of Electricity by the rubbing movement of backpack when attached to the piezoelectric sensors. This project proves that if it is used correctly then the harvested energy produced can be much higher than that electro-active system itself.

A similar research explores that energy can be harvested by the bobbing motion of a backpack produced during walking or running, thus, converting the whole body motion into a power source, allowing the backpack to move up and down and converting this linear motion of backpack into the electrical motion. Researchers concluded that around 4W to 7.5W of energy can be generated by this bobbing motion of backpack which can be used to power many devices simultaneously.

Military in this manner have gone so far that it has modified commercial technologies to energy harvesting. Hand cranks have always been used to power small devices in faraway areas where onsite energy is required. Hence, the need of onsite energy is still apparent and small scale energy harvesting systems are considered as energy harvesters.

Consequently, it's seen that man power build-up many purposes in contemporary society. This is beneficial when operators are lonely with natural disaster, or are in a faraway discipline. This supplies for a natural, convenient for enforce also moderately little fee project which is certainly valuable in countryside areas of setting up countries because skill in working equipment and funding budget is constrained. Achievement of energy via no adequate man energy can also be viable because it may be beneficial for more than a few novel portable electronics functions. Additionally, it might allow for energy new release to be accomplished socially, taking out the sense of adequate energy whilst growing the energy production meaningfully. The notion of making use of man power as a substitute and regenerative power foundation is attaining repute to the near that companies have made round changing activity gear comparable to stationary bikes and elliptical to electrical power generators.

3. PROBLEM STATEMENT AND MAIN CONTRIBUTION

The sector's energy consumption is all time high with the claim constantly growing. The trouble cause many irreversible problems that have got to be spoken and taken care of.

• Reduction of sources due to finite accessibility of non-regenerative energy sources, e.g. bio gas
• Air pollution, e.g. Firewood usage in vigor crops
• Growing population, specifically in constructing international locations with deficiency of fresh energy.
• Environmental pollution related with local weather variations and hostile implications

These cause a lot dispute within the developed countries; nonetheless, past results have additionally exposed a much more normal undertaking of accessibility in the less developed countries.

Information got by the arena bank acquired just lately in 2014 appraised that about 34.9% sectors population have no reach of electrical power. Many incorporate that have very constrained availability to electrical energy. Additional, many countries with the cheapest prices for part of population through electrical power even have cheap prices of city people percentage.

4. PROBLEM SOLUTION

The most feasible way of energy harvesting is during exercising using machines. The actions performed on these exercising machines help to burn calories and to improve muscle movement and shape with repetitive motions and resistive movements. The main purpose of these machines is to provide certain motions and resistances to the human body again and again respectively. The desire of these repetitive motions and resistance can result in producing electricity when these machines are connected to certain power generating devices [7]. So, the mechanical power exerted on these exercising machines can be harvested and converted into electricity and can even be stored [8][9].

The main and the most important exercising machine that can be used in electrical energy harvesting is the exercising bike which can be seen in most of the gyms for the physical fitness. These bicycles have a feasible rotatory system, thus, harnessing the cyclic and rotatory motion of the cycle into the electrical energy [10]. This machine has much technical feasibility of harvesting human energy as compared to other machines and is considered economically viable.

4.1 Methodology

In first view, we have to design a prototype model of each of the equipment attached to a compact generator that converts the motion of the wheels or the mechanical pulley attached to each equipment into electricity, which is then fed into the power grid, offsetting some of the energy use. For these gym-goers, it's not only just about their cardio fitness; their exercise is making the planet-greener**.**

The produced energy of the machine is sent either directly to the battery or is consumed by the appliances connected to it. The power produced is then checked by using the android app which communicated with the machine getting the information about the force applied on the machine and the energy produced by it.

This project provides you an efficient and yet controllable power source with an efficient storage and distribution system preserving the environment by producing environment friendly power sources by a process shown in Figure 4.1. The project harvests the mechanical energy into electrical energy by using a generator-based system .The shaft of the generator will be attached to the exercising equipment thus produced electrical energy will be used for powering up a small appliance or it will be stored directly into battery.

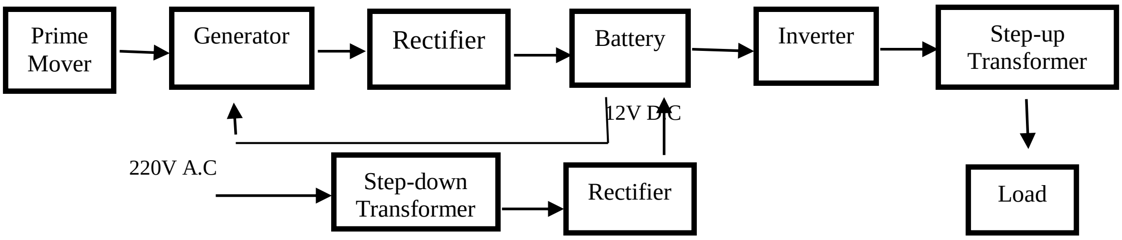

Figure 4.1: Block Diagram

Figure 4.2(a): Elliptical Exercise Cycle attached to Dynamo; Figure 4.2(b): Manual Treadmill attached to Dynamo

Figure 4.3: Dynamo

5. Conclusion

We constructed a Smart Energy Harvesting system using physical exercising machines which are feasible enough to harvest power and seem to generate a source of renewable energy. This biomechanical energy harvesting constitutes a clean, portable energy alternative to the fossil fuels currently being used. The energy expended in a typical workout at the gym is usually wasted in the mechanics of the equipment. This project harvests the mechanical energy of the machine and converts it to electrical energy using a generator based system. The exercise equipment will be attached to a dynamo. This creates a DC voltage which is fed into a DC-DC converter and then sent to a battery where it can be stored for future use. This proposed system will help consuming our physical energy in some useful way.

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